Simple Function Definitions

This file demonstrates basic function definitions in Z notation that work well with fuzz type checking.

Example 1 : Simple Total Functions

Total functions are defined over their entire domain.

$$\begin{array}{|l} square : \mathbb{N} \longrightarrow \mathbb{N} \\ \hline \forall n : \mathbb{N} \bullet square(n) = n * n \end{array}$$

$$\begin{array}{|l} successor : \mathbb{N} \longrightarrow \mathbb{N} \\ \hline \forall n : \mathbb{N} \bullet successor(n) = n + 1 \end{array}$$

$$\begin{array}{|l} double : \mathbb{N} \longrightarrow \mathbb{N} \\ \hline \forall n : \mathbb{N} \bullet double(n) = 2 * n \end{array}$$

Example 2 : Partial Functions

Partial functions are not defined over their entire domain.

$$\begin{array}{|l} predecessor : \mathbb{N} \nrightarrow \mathbb{N} \\ \hline \forall n : \mathbb{N} \mid n > 0 \bullet predecessor(n) = n - 1 \end{array}$$

predecessor is partial on natural numbers since 0 has no predecessor in N. The bullet separator filters to positive numbers (where predecessor is defined), then specifies the function value.

Example 3 : Generic Functions

Generic functions work with any type parameter.

$$\begin{array}{|l} [X] \\ identity : X \longrightarrow X \\ \hline \forall x : X \bullet identity(x) = x \end{array}$$

$$\begin{array}{|l} [X, Y] \\ fst : X \times Y \longrightarrow X \\ \hline \forall x : X ; y : Y \bullet fst(x, y) = x \end{array}$$

$$\begin{array}{|l} [X, Y] \\ snd : X \times Y \longrightarrow Y \\ \hline \forall x : X ; y : Y \bullet snd(x, y) = y \end{array}$$

Example 4 : Functions with Given Types

$$[Person, Department]$$

$$\begin{array}{|l} assignment : Person \nrightarrow Department \end{array}$$

assignment is a partial function from Person to Department.

Example 5 : Functions on Numbers

$$
\begin{array}{|l}
triple : \mathbb{N} \longrightarrow \mathbb{N} \\
\hline
\forall\, n : \mathbb{N} \bullet triple(n) = 3 * n
\end{array}
$$

$$
\begin{array}{|l}
addOne : \mathbb{Z} \longrightarrow \mathbb{Z} \\
\hline
\forall\, n : \mathbb{Z} \bullet addOne(n) = n + 1
\end{array}
$$

Example 6 : Function with Given Types

$[Student, Grade]$

$$
\begin{array}{|l}
grades : Student \nrightarrow Grade
\end{array}
$$

grades maps students to their grades.

Example 7 : Function Composition Example

$$
\begin{array}{|l}
f : \mathbb{N} \longrightarrow \mathbb{N} \\
g : \mathbb{N} \longrightarrow \mathbb{N} \\
h : \mathbb{N} \longrightarrow \mathbb{N} \\
\hline
\forall\, n : \mathbb{N} \bullet f(n) = 2 * n \\
\forall\, n : \mathbb{N} \bullet g(n) = n + 1 \\
h = f \fatsemi g
\end{array}
$$

f doubles its input, g adds one to its input. h is their forward composition: h(n) = f(g(n)) = 2 * (n + 1).

Example 8 : Modulo Function

$$
\begin{array}{|l}
modulo3 : \mathbb{N} \longrightarrow \mathbb{N} \\
\hline
\forall\, n : \mathbb{N} \bullet modulo3(n) = n \mathsf{\ mod\ } 3
\end{array}
$$

For comprehensive function examples including recursive functions, higher-order functions, and advanced patterns, see examples/09_sequences/pattern_matching.txt.